MIL-B-36162C
3 November 1987
SUPERSEDING
MIL-B-36162B
9 August 1978

MILITARY SPECIFICATION

BLADES, LARYNGOSCOPE, MILLER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers detachable and hook-on laryngoscope blades of the Miller design, suitable for use with battery handles of laryngoscopes.

1.2 Classification. The laryngoscope blades shall be of the following sizes, as specified (see 6.2):

Size 0	-	Premature Infant
Size 1	-	Infant
Size 2	-	Medium Adult
Size 3	-	Large Adult

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-RST, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

GG-I-526	- Instruments, Dental and Surgical; General Specifications.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-L-10547	- Liners, Case and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.
MIL-L-36628	- Laryngoscope Set, MacIntosh.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Parts and Equipment, Procedures for Packaging of.

2.1.2 Drawing. The following drawing forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

DRAWING

Defense Personnel Support Center (DPSC)

24141	- BLADES, LARYNGOSCOPE, MILLER

(Copies of specifications, standards and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials, Tests for.

E 92 - Vickers Hardness of Metallic Materials, Tests for.

E 140 - Hardness Conversion Tables for Metals (Relationship Between Brinell Hardness, Vickers Hardness, Rockwell Hardness, Rockwell Superficial Hardness and Knoop Hardness).

F 899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments.

F 965 - Rigid Laryngoscopes for Tracheal Intubation.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The laryngoscope blades shall be fabricated from materials specified in MIL-L-36628 and ASTM F899. The Rockwell hardness of the Class 3 material shall be B75 minimum when tested as specified in 4.4.3. The material of the tubing from the lock to the bulb holder shall be corrosion-resistant stainless steel, properly annealed to permit bending.

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3.2 Style, design and dimensions. Shall be a laryngoscope blade of the Miller design, suitable for use with universal large battery type handle, specified in MIL-L-36628. Blades shall be of the detachable and hook-on type, suitable for the introduction of endotracheal tubes in the administration of anesthesia. The blades shall be capable of insertion onto the battery handle specified above and designed to light at the distal end. The hook-on fitting shall be in accordance with ASTM Standard F965. A frosted lamp, in accordance with MIL-L-36628 for Size 0 and 1 Type I lamps, and Size 2 and 3 Type II lamp, shall be assembled to the blade. Blades shall be in strict accordance with DPSC Drawing 24141 for appropriate size and National Stock Number (NSN).

3.2.1 Internal threads of lamp socket. Internal threads of the lamp socket shall allow the "GO" thread plug gages to be inserted and turn freely, and shall not allow entry of the "NO GO" thread gage when tested as specified in 4.4.2.

3.3 Finish. Finish on all edges and surfaces shall be uniform and free from burrs, sharp edges, pits, crevices, grind marks, rough areas and cracks. Finish shall be mirror and satin as indicated on DPSC Drawing 24141.

3.4 Resistance to boil test and salt-spray test. The finished blades shall show no evidence of corrosion when subjected to the boil test specified in 4.4.1.1, and the salt-spray test specified in 4.4.1.2.

3.4.1 Interchangeability and illumination test. The blade hook-on fitting shall be interchangeable with a battery handle that is in accordance with the hook-on fitting specified in ASTM F965. Further, the lamp shall illuminate, provided that the handle contains new batteries and the blade has a new lamp.

3.5 Identification markings. Each blade shall be permanently and legibly marked with the contractor's name or registered trademark (and country of origin when manufactured in countries outside of the United States). Blades shall be permanently and legibly marked "MILLER 0" or "MILLER 1", as applicable. Laryngoscope blades manufactured from Class 3 material shall also be marked "stainless".

3.6 Workmanship. Workmanship shall be first class throughout. The blades shall be free from defects which detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection. Unless otherwise specified, inspection shall be in accordance with GG-I-526.

4.1.3 Quality conformance inspection. Certificates of quality, supplied by the manufacturer of the corrosion-resisting steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity, has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSNs, the item identification, the name of the material, the lot number, the lost size, the sample size, the date of testing, the test method, individual test results, and the the specification requirements.

4.1.4 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table I. Unit of product for examination purposes shall be one blade.

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TABLE I. Sampling criteria for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	I	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as indicated in Table II. The unit of product for test purposes shall be one laryngoscope blade. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

TABLE II. Sampling for tests.

Component	Character-istic	Require-ment	Test procedure	Inspection level	AQL (percent defective)
Blade	Dimensions	3.2	-	S-2	2.5
	Internal thread test	3.2.1	4.4.2	S-2	2.5
	Plating test	3.1.1, and GG-I-526	GG-I-526	S-2	2.5
	Boil test	3.4	4.4.1.1	S-2	2.5
	Salt-spray test	3.4	4.4.1.2	S-2	2.5
	Hardness (for class 3 material)	3.1.1	4.4.3	S-2	2.5
Lamp	For all requirements, see MIL-L-36628 for Type I lamps.				
Blade & Handle	inter-changeability	3.4.1 and ASTM F965	ASTM F965	___ *	___ *
	Illumination	3.4.1 and ASTM F965	ASTM F965	___ *	___ *

* Note: Defects are not permissable.

4.2.3. Raw materials. A sufficient quantity of each lot of metal used in the manufacture of the laryngoscope blades shall be selected for chemical analysis.

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4.3 Examination.

4.3.1 Examination of blades. The laryngoscope blades shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any blades in the sample containing one or more defects shall be rejected and if the number of defective blades in any sample exceeds the acceptance number for the sample, the lot represented by that sample shall be rejected.

TABLE III. Classification of defects.

Categories	Defects
Critical	None defined
Major	
101	Blades not of the specified size.
102	Blades not of detachable, hook-on design.
103	Blades do not positively seat (hook-on) when inserted onto a universal battery handle in accordance with MIL-L-36628.
104	Blades not of finish specified.
105	Blades not free from pits, cracks, crevices, sharp edges, burrs, scale or corrosion.
106	Internal threads of lamp socket and external threads of lamps not complete or not well defined.
107	Glass envelope of lamp not properly secured to lamp base.
108	Glass envelope of lamp cracked.
109	Plated parts not free from flaking.
110	Blades not permanently or not legibly marked as specified in 3.5.
Minor	
201	Glass envelope of lamp not free from bubbles, dirt or other imperfections.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Corrosion tests.

4.4.1.1 Boil test. Laryngoscope blades shall be subjected to the boil test as specified in GG-I-526.

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4.4.1.2 Salt-spray test. Laryngoscope blades shall be subjected to the salt-spray test as specified in GG-I-526.

4.4.2 Internal thread test of lamp socket. The contractor shall provide the necessary thread plug gages to the Government Quality Assurance Representative (QAR) for checking the internal threads of the lamp socket in accordance with the thread dimensions shown on DPSC Drawing 24141. The following dimensional parameters shall be inscribed on the gages.

For size 0 and 1

Thread size: 1/8-72NS-3
"GO" PITCH DIAMETER - 0.1160
"NO GO" PITCH DIAMETER - 0.1174

For size 2 and 3

Thread size: 8-32UNC-2B
"GO" PITCH DIAMETER - 0.1437
"NO GO" PITCH DIAMETER - 0.1475

To perform thread tests of lamp socket, remove lamp from lamp socket and place "GO" plug gage into lamp socket. Remove "GO" gage and place "NO GO" plug gage into lamp socket. Fits of gages into lamp sockets shall be noted.

4.4.3 Hardness test. Hardness shall be determined in accordance with ASTM Standards E 18 or E 92. Conversion of Vickers hardness values to Rockwell values shall be in accordance with ASTM Standard E 140.

4.5 Inspection of packaging. The sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. Blades shall be cleaned and dried in accordance with MIL-P-116.

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5.1.1.2 Preservative applications. Preservatives shall not be used.

5.1.1.3 Unit package. Each blade shall be packaged in accordance with MIL-P-116, Method IC-1.

5.1.1.4 Intermediate package. Twelve unit packages containing blades of one size only shall be packaged in a box conforming to PPP-B-566 or PPP-B-636, class domestic or PPP-B-676. Closure shall be as specified in the applicable box specification.

5.1.2 Level C.

5.1.2.1 Unit package. Preservation for blades shall conform to the requirements of MIL-STD-794 for level C.

5.1.2.2 Intermediate package. Twelve unit packages containing blades of one size only shall be packaged in accordance with the requirements of MIL-STD-794 for level C.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Twenty-four intermediate packages (288 units) of size "0" blades or twelve intermediate packages (144 units) of one size only of blade sizes 1 thru 4 shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type, PPP-B-621, class 2, or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Twenty-four intermediate packages (288 units) of size "0" blades or twelve intermediate packages (144 units) of one size only of blades sizes 1 thru 4 shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic type, PPP-B-621, class 1 or PPP-B-636 class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. Twenty-four intermediate packages (288 units) of size "0" blades or twelve intermediate packages (144 units) of one size only of blades sizes 1 thru 4 shall be packed in accordance with the requirements of MIL-STD-794 for level C.

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5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination exceed 250 pounds or 20 cubic feet. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 20 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 54 inches in height, 48 inches in width, and 40 inches in length. Pallet shall conform to type IV or type V.

5.2.5.1 Levels A and B. Blades packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Blades packed as specified in 5.2.3 shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C. Each unit pack, intermediate package, exterior container and unitized load shall be marked as specified in MIL-STD-129.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.5.

6. NOTES

6.1 Intended use. The laryngoscope blades are used in medical procedures concerning the human larynx.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) National Stock Number.

6.3 National Stock Number coverage. This specification covers the following items:

National Stock No.	Item Identification
6515-00-656-0474	BLADE, LARYNGOSCOPE, Infant, Miller
6515-00-656-0475	BLADE, LARYNGOSCOPE, Premature Infant, Miller
6515-01-124-9608	BLADE, LARYNGOSCOPE, Medium Adult, Miller
6515-01-125-6615	BLADE, LARYNGOSCOPE, Adult, Miller

6.4 Subject term (key word) listing.

Blades
Laryngoscope blades
Miller laryngoscope blades

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6515-3831

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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